



Figure 4: A shear transformation

Linear Transformations

Definition: A transformation (or mapping) T is linear if:

- (i) $T(u+v) = T(u) + T(v)$ for all u, v in the domain of T ;
- (ii) $T(cu) = cT(u)$ for all scalars c and all u in the domain of T .

By Theorem 5 in Section 1.4, Every matrix transformation is a linear transformation.

- Linear transformations preserve the operations of vector addition and scalar multiplication.
- Property (i) says that the result $T(u+v)$ of first adding u and v in $\mathbb{R}^n$ and then applying T is the same as first applying T to u and to v and then adding $T(u)$ and $T(v)$ in $\mathbb{R}^m$. These two properties lead easily to the following useful facts.

If T is a linear transformation, then $T(0) = 0$ (3) and $T(cu+dv) = cT(u) + dT(v)$ (4) for all vectors u, v in the domain of T and all scalars c, d .

Proof: (3). $T(0) = T(0 \cdot 0) = 0 \cdot T(0)$ (condn. (ii) from defn.) $= 0$
 (4) $T(cu+dv) = T(cu) + T(dv)$ (condn. (i) from defn.) $= cT(u) + dT(v)$ (condn. (ii) from defn.)

Observe that if a transformation satisfies (4) for all u, v and c, d , it must be linear. (Set $c=d=1$ for preservation of addition, and set $d=0$ for preservation of scalar multiplication.)

$$T(c_1v_1 + \dots + c_pv_p) = c_1T(v_1) + \dots + c_pT(v_p) \quad (5)$$

In engineering and physics, (5) is referred to as a superposition principle. Think of $v_1, \dots, v_p$ as signals that go into a system and $T(v_1), \dots, T(v_p)$ as the responses of that system to the signals.